

BRANCH-SW-07

cisco_ios · 12.4

0 PASSED	48 FAILED	4 NEEDS REVIEW
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52 control result(s) across 4 framework(s). A control evaluated under two frameworks is counted once per framework.

Compliance Summary

FRAMEWORK	PASS	FAIL	REVIEW	TOTAL	SCORE	ADJUDICATED
CIS	0	12	1	13	0.0%	0.0%
NIST SP 800-53	0	12	1	13	0.0%	0.0%
DISA STIG	0	12	1	13	0.0%	0.0%
ISO/IEC 27001	0	12	1	13	0.0%	0.0%

Device Identification

Hostname	BRANCH-SW-07	L8: hostname BRANCH-SW-07
Vendor	cisco_ios	detected by parser selection
OS family	ios	declared by the parser
OS version	12.4	L4: version 12.4
Model	null	Hardware model is not present in a cisco ios configuration file; it requires show-command output (show version / show inventory).
Serial number	null	Serial number is not present in a cisco ios configuration file; it requires show-command output (show version / show inventory).

A configuration file does not carry the hardware serial number. Fields shown as null were not present in the ingested file and were not inferred; the evidence column names the show command that would establish them.

Source and Provenance

Source file	samples/insecure_ios.conf	
SHA-256	4202c7c96860475b89031bb4f8f5e45dbb79c541e5f163f33cd2faaf786df175	of the ingested bytes
Ingested	2026-08-23 19:23:38 UTC	
Status	audited	
Parser	cisco_ios v1.0.0	detection confidence 1.00
Config lines	48	
Parser notes	2 warning(s)	2 'line vty' block(s) have no inbound 'access-class', so remote management is reachable from any source address.; No 'ip ssh version' statement found; IOS may fall back to version 1.99 (v1 compatibility) depending on release and RSA key state.

Control Results

FRAMEWORK	CONTROL	SEV	RESULT	TITLE	EVIDENCE
CIS	1.2	high	FAIL	Restrict management access by source IP	<absent> 2 'line vty' block(s) have no inbound 'access-class', so remote management is reachable from any source address.
CIS	1.2.2	high	FAIL	Restrict VTY transport to secure protocols only	L41: transport input telnet
CIS	1.4.1, 1.4.2	high	FAIL	Store privileged credential as hash and encrypt passwords	<absent> No 'enable secret' statement present. IOS always writes this command to the running-config when configured, so its absence means no enable secret is set.
CIS	1.5	high	FAIL	Do not configure read-write SNMP communities	L27: snmp-server community public R0
CIS	1.5.2, 1.5.3	high	FAIL	Unset default SNMP community names	L27: snmp-server community public R0
CIS	1.1	medium	FAIL	Enforce minimum password length	<absent> No 'security passwords min-length' statement present. IOS enforces no minimum password length unless this is configured, and writes it when it is.
CIS	1.1.1	medium	FAIL	Enable AAA Authentication	<absent> No 'aaa new-model' statement present. IOS writes this command to the running-config when enabled, so its absence means AAA is not enabled.
CIS	1.2.9	medium	FAIL	Set VTY session idle timeout	L40: exec-timeout 0 0
CIS	2.1	medium	FAIL	Disable HTTP management server	L14: ip http server
CIS	2.2.2, 2.2.4	medium	FAIL	Configure a logging destination	<absent> No 'logging host' or 'logging buffered' statement present. Both are written to the running-config when configured, so absence means no log destination exists.
CIS	2.3	medium	FAIL	Configure at least one NTP server	<absent> No 'ntp server' statement present. IOS writes NTP configuration to the running-config, so absence means the clock is not synchronised.
CIS	1.6	low	FAIL	Configure a login banner	<absent> No 'banner login', 'banner motd' or 'banner exec' statement present. IOS writes banner configuration to the running-config, so absence means no banner is shown.
CIS	2.1.1.6	high	REVIEW	Enforce SSH protocol version 2	<no evidence> No 'ip ssh version' statement found; IOS may fall back to version 1.99 (v1 compatibility) depending on release and RSA key state.
NIST SP 800-53	no_default_snmp_community *	high	FAIL	Unset default SNMP community names	L27: snmp-server community public R0

FRAMEWORK	CONTROL	SEV	RESULT	TITLE	EVIDENCE
NIST SP 800-53	no_write_snmp_community *	high	FAIL	Do not configure read-write SNMP communities	L27: snmp-server community public R0
NIST SP 800-53	AC-17	high	FAIL	Restrict VTY transport to secure protocols only	L41: transport input telnet
NIST SP 800-53	AC-3	high	FAIL	Restrict management access by source IP	<absent> 2 'line vty' block(s) have no inbound 'access-class', so remote management is reachable from any source address.
NIST SP 800-53	IA-5	high	FAIL	Store privileged credential as hash and encrypt passwords	<absent> No 'enable secret' statement present. IOS always writes this command to the running-config when configured, so its absence means no enable secret is set.
NIST SP 800-53	AC-12	medium	FAIL	Set VTY session idle timeout	L40: exec-timeout 0 0
NIST SP 800-53	AC-2	medium	FAIL	Enable AAA Authentication	<absent> No 'aaa new-model' statement present. IOS writes this command to the running-config when enabled, so its absence means AAA is not enabled.
NIST SP 800-53	AU-2	medium	FAIL	Configure a logging destination	<absent> No 'logging host' or 'logging buffered' statement present. Both are written to the running-config when configured, so absence means no log destination exists.
NIST SP 800-53	AU-8	medium	FAIL	Configure at least one NTP server	<absent> No 'ntp server' statement present. IOS writes NTP configuration to the running-config, so absence means the clock is not synchronised.
NIST SP 800-53	IA-5(1)	medium	FAIL	Enforce minimum password length	<absent> No 'security passwords min-length' statement present. IOS enforces no minimum password length unless this is configured, and writes it when it is.
NIST SP 800-53	SC-7	medium	FAIL	Disable HTTP management server	L14: ip http server
NIST SP 800-53	AC-8	low	FAIL	Configure a login banner	<absent> No 'banner login', 'banner motd' or 'banner exec' statement present. IOS writes banner configuration to the running-config, so absence means no banner is shown.
NIST SP 800-53	SC-13	high	REVIEW	Enforce SSH protocol version 2	<no evidence> No 'ip ssh version' statement found; IOS may fall back to version 1.99 (v1 compatibility) depending on release and RSA key state.
DISA STIG	management_acl *	high	FAIL	Restrict management access by source IP	<absent> 2 'line vty' block(s) have no inbound 'access-class', so remote management is reachable from any source address.

FRAMEWORK	CONTROL	SEV	RESULT	TITLE	EVIDENCE
DISA STIG	no_default_snmp_community *	high	FAIL	Unset default SNMP community names	L27: snmp-server community public R0
DISA STIG	no_write_snmp_community *	high	FAIL	Do not configure read-write SNMP communities	L27: snmp-server community public R0
DISA STIG	CCI-000200	high	FAIL	Store privileged credential as hash and encrypt passwords	<absent> No 'enable secret' statement present. IOS always writes this command to the running-config when configured, so its absence means no enable secret is set.
DISA STIG	CCI-000366	high	FAIL	Restrict VTY transport to secure protocols only	L41: transport input telnet
DISA STIG	CCI-000015	medium	FAIL	Enable AAA Authentication	<absent> No 'aaa new-model' statement present. IOS writes this command to the running-config when enabled, so its absence means AAA is not enabled.
DISA STIG	CCI-000057	medium	FAIL	Set VTY session idle timeout	L40: exec-timeout 0 0
DISA STIG	CCI-000130	medium	FAIL	Configure a logging destination	<absent> No 'logging host' or 'logging buffered' statement present. Both are written to the running-config when configured, so absence means no log destination exists.
DISA STIG	CCI-000159	medium	FAIL	Configure at least one NTP server	<absent> No 'ntp server' statement present. IOS writes NTP configuration to the running-config, so absence means the clock is not synchronised.
DISA STIG	CCI-000200	medium	FAIL	Enforce minimum password length	<absent> No 'security passwords min-length' statement present. IOS enforces no minimum password length unless this is configured, and writes it when it is.
DISA STIG	CCI-000381	medium	FAIL	Disable HTTP management server	L14: ip http server
DISA STIG	CCI-000048	low	FAIL	Configure a login banner	<absent> No 'banner login', 'banner motd' or 'banner exec' statement present. IOS writes banner configuration to the running-config, so absence means no banner is shown.
DISA STIG	CCI-000068	high	REVIEW	Enforce SSH protocol version 2	<no evidence> No 'ip ssh version' statement found; IOS may fall back to version 1.99 (v1 compatibility) depending on release and RSA key state.
ISO/IEC 27001	enable_secret_encrypted *	high	FAIL	Store privileged credential as hash and encrypt passwords	<absent> No 'enable secret' statement present. IOS always writes this command to the running-config when configured, so its absence means no enable secret is set.

FRAMEWORK	CONTROL	SEV	RESULT	TITLE	EVIDENCE
ISO/IEC 27001	management_acl *	high	FAIL	Restrict management access by source IP	<absent> 2 'line vty' block(s) have no inbound 'access-class', so remote management is reachable from any source address.
ISO/IEC 27001	no_default_snmp_community *	high	FAIL	Unset default SNMP community names	L27: snmp-server community public R0
ISO/IEC 27001	no_write_snmp_community *	high	FAIL	Do not configure read-write SNMP communities	L27: snmp-server community public R0
ISO/IEC 27001	A.8.20	high	FAIL	Restrict VTY transport to secure protocols only	L41: transport input telnet
ISO/IEC 27001	ntp_configured *	medium	FAIL	Configure at least one NTP server	<absent> No 'ntp server' statement present. IOS writes NTP configuration to the running-config, so absence means the clock is not synchronised.
ISO/IEC 27001	http_server_disabled *	medium	FAIL	Disable HTTP management server	L14: ip http server
ISO/IEC 27001	A.5.17	medium	FAIL	Enforce minimum password length	<absent> No 'security passwords min-length' statement present. IOS enforces no minimum password length unless this is configured, and writes it when it is.
ISO/IEC 27001	A.8.10	medium	FAIL	Configure a logging destination	<absent> No 'logging host' or 'logging buffered' statement present. Both are written to the running-config when configured, so absence means no log destination exists.
ISO/IEC 27001	A.8.19	medium	FAIL	Set VTY session idle timeout	L40: exec-timeout 0 0
ISO/IEC 27001	A.8.2	medium	FAIL	Enable AAA Authentication	<absent> No 'aaa new-model' statement present. IOS writes this command to the running-config when enabled, so its absence means AAA is not enabled.
ISO/IEC 27001	login_banner *	low	FAIL	Configure a login banner	<absent> No 'banner login', 'banner motd' or 'banner exec' statement present. IOS writes banner configuration to the running-config, so absence means no banner is shown.
ISO/IEC 27001	ssh_version_2 *	high	REVIEW	Enforce SSH protocol version 2	<no evidence> No 'ip ssh version' statement found; IOS may fall back to version 1.99 (v1 compatibility) depending on release and RSA key state.

* Internal semantic mapping, not an official framework clause number. The control intent is the framework's; the identifier is this tool's, because the clause could not be verified against a licensed copy of the benchmark.

Detailed Findings

FAIL [HIGH] CIS-IOS-1.2-VTY-ACCESS-CLASS — Restrict management access by source IP

Framework: CIS · Control: 1.2 (verified reference)

An ACL/filter must be applied to restrict administrative access to authorized source subnets.

Why: management_acl_applied=False (required: is_true)

Evidence

management_acl_applied	<absent> 2 'line vty' block(s) have no inbound 'access-class', so remote management is reachable from any source address.
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Remediation: Permit only the management network, and apply the list inbound on every VTY line.

```
configure terminal
access-list 99 remark MANAGEMENT-HOSTS
access-list 99 permit 10.20.30.0 0.0.0.255
line vty 0 15
access-class 99 in
end
copy running-config startup-config
```

FAIL [HIGH] CIS-IOS-1.2.2 — Restrict VTY transport to secure protocols only

Framework: CIS · Control: 1.2.2 (verified reference)

VTY lines must accept SSH only. Telnet and other legacy cleartext transports must be disabled.

Why: telnet_enabled=True (required: is_false); vty_transport_input=['acercon', 'lat', 'pad', 'rlogin', 'ssh', 'telnet', 'udptn'] (required: subset_of ['ssh'])

Evidence

telnet_enabled	L41: transport input telnet
vtty_transport_input	L41: transport input telnet

Remediation: Restrict every VTY line to the SSH transport.

```
configure terminal
line vty 0 15
transport input ssh
end
copy running-config startup-config
```

FAIL [HIGH] CIS-IOS-1.4.1-1.4.2 — Store privileged credential as hash and encrypt passwords

Framework: CIS · Control: 1.4.1, 1.4.2 (verified reference)

The privileged credential must be stored as a one-way hash, and stored passwords elsewhere in the configuration must not be left in plaintext.

Why: enable_secret_set=False (required: is_true); password_encryption=False (required: is_true)

Evidence

enable_secret_set	<absent> No 'enable secret' statement present. IOS always writes this command to the running-config when configured, so its absence means no enable secret is set.
password_encryption	<absent> No 'service password-encryption' statement present. The feature is disabled by default and is written to the running-config when enabled, so absence means disabled.

Remediation: Configure a hashed enable secret and turn on password encryption; remove the legacy enable password.

```
configure terminal
enable secret <strong-password>
no enable password
service password-encryption
end
copy running-config startup-config
```

FAIL [HIGH] CIS-IOS-1.5-SNMP-NO-WRITE — Do not configure read-write SNMP communities

Framework: CIS · Control: 1.5 (verified reference)

No SNMP v1/v2c community may grant read-write access.

Why: snmp_communities.access=['ro', 'rw'] (required: contains_none ['rw'])

Evidence

snmp_communities	L27: snmp-server community public R0
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Remediation: Remove every read-write community; use read-only with an ACL, or SNMPv3 for write access.

```
configure terminal
no snmp-server community <rw-string> RW
snmp-server community <new-ro-string> R0 99
end
copy running-config startup-config
```

FAIL [HIGH] CIS-IOS-1.5.2-1.5.3 — Unset default SNMP community names

Framework: CIS · Control: 1.5.2, 1.5.3 (verified reference)

No SNMP community string may use a well-known default value such as 'public' or 'private'.

Why: snmp_communities.name=['public', 'private'] (required: contains_none ['public', 'private'])

Evidence

snmp_communities	L27: snmp-server community public R0
------------------	--------------------------------------

Remediation: Remove default community strings and replace them with unique, non-guessable values restricted by an ACL.

```
configure terminal
no snmp-server community public
no snmp-server community private
access-list 99 permit <nms-subnet> <wildcard>
snmp-server community <unique-string> R0 99
end
copy running-config startup-config
```

FAIL [MEDIUM] CIS-IOS-1.1-PASSWORD-MIN-LENGTH — Enforce minimum password length

Framework: CIS · Control: 1.1 (verified reference)

The device must refuse locally configured passwords shorter than eight characters.

Why: password_min_length=0 (required: greater_or_equal 8)

Evidence

password_min_length	<absent> No 'security passwords min-length' statement present. IOS enforces no minimum password length unless this is configured, and writes it when it is.
---------------------	---

Remediation: Enforce a minimum length for every locally configured password.

```
configure terminal
security passwords min-length 8
end
copy running-config startup-config
```

FAIL [MEDIUM] CIS-IOS-1.1.1 — Enable AAA Authentication

Framework: CIS · Control: 1.1.1 (verified reference)

AAA must be enabled so that authentication, authorization and accounting for device access are centrally defined rather than falling back to line-local passwords.

Why: aaa_enabled=False (required: is_true)

Evidence

aaa_enabled	<absent> No 'aaa new-model' statement present. IOS writes this command to the running-config when enabled, so its absence means AAA is not enabled.
-------------	---

Remediation: Enable the AAA subsystem, then define authentication methods before disconnecting.

```
configure terminal
aaa new-model
end
copy running-config startup-config
```

FAIL [MEDIUM] CIS-IOS-1.2.9 — Set VTY session idle timeout

Framework: CIS · Control: 1.2.9 (verified reference)

Idle VTY sessions must be torn down automatically within 10 minutes (600 seconds) or less.

Why: vty_exec_timeout_seconds=0 (required: greater_than 0)

Evidence

vtty_exec_timeout_seconds	L40: exec-timeout 0 0
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Remediation: Set an idle timeout of 10 minutes or less on every VTY line.

```
configure terminal
line vty 0 15
exec-timeout 10 0
end
copy running-config startup-config
```

FAIL [MEDIUM] CIS-IOS-2.1-HTTP-SERVER — Disable HTTP management server

Framework: CIS · Control: 2.1 (verified reference)

The on-box unencrypted HTTP management server must be disabled.

Why: http_server_enabled=True (required: is_false)

Evidence

http_server_enabled	L14: ip http server
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Remediation: Disable the HTTP server; use HTTPS or SSH if a management interface is genuinely required.

```
configure terminal
no ip http server
no ip http secure-server
end
copy running-config startup-config
```

FAIL [MEDIUM] CIS-IOS-2.2.2-2.2.4 — Configure a logging destination

Framework: CIS · Control: 2.2.2, 2.2.4 (verified reference)

Logging must be enabled and must have at least one destination: a remote syslog host, a sized local buffer, or both.

Why: logging_enabled=False (required: is_true); logging_hosts=[] (required: is_not_empty); logging_buffered=False (required: is_true)

Evidence

logging_enabled	<absent> No 'logging host' or 'logging buffered' statement present. Both are written to the running-config when configured, so absence means no log destination exists.
logging_hosts	<absent> No 'logging host' statement present; no remote syslog destination is configured.
logging_buffered	<absent> No 'logging buffered' statement present; local log buffering is not configured.

Remediation: Enable logging to a sized local buffer and to a remote syslog collector.

```
configure terminal
logging on
logging buffered 64000 informational
logging host <syslog-server-ip>
logging trap informational
service timestamps log datetime msec show-timezone
end
copy running-config startup-config
```


FAIL [MEDIUM] CIS-IOS-2.3-NTP-CONFIGURED — Configure at least one NTP server

Framework: CIS · Control: 2.3 (verified reference)

The device must synchronise its clock with at least one NTP time source.

Why: ntp_servers=[] (required: is_not_empty)

Evidence

ntp_servers	<absent> No 'ntp server' statement present. IOS writes NTP configuration to the running-config, so absence means the clock is not synchronised.
-------------	---

Remediation: Point the device at two or more internal time sources.

```
configure terminal
ntp server 10.20.30.41
ntp server 10.20.30.42
clock timezone UTC 0 0
end
copy running-config startup-config
```

FAIL [LOW] CIS-IOS-1.6-LOGIN-BANNER — Configure a login banner

Framework: CIS · Control: 1.6 (verified reference)

A banner must be displayed before or at login to warn unauthorized users.

Why: login_banner_present=False (required: is_true)

Evidence

login_banner_present	<absent> No 'banner login', 'banner motd' or 'banner exec' statement present. IOS writes banner configuration to the running-config, so absence means no banner is shown.
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Remediation: Set a login banner stating that access is restricted and activity is monitored.

```
configure terminal
banner login ^C
Authorised access only. Activity is logged and monitored.
^C
end
copy running-config startup-config
```

NEEDS_REVIEW [HIGH] CIS-IOS-2.1.1.6 — Enforce SSH protocol version 2

Framework: CIS · Control: 2.1.1.6 (verified reference)

The device must enforce SSH protocol version 2 for secure administrative management.

Why: No conclusive evidence for ssh_version. No 'ip ssh version' statement found; IOS may fall back to version 1.99 (v1 compatibility) depending on release and RSA key state.

Evidence

ssh_version	<no evidence> No 'ip ssh version' statement found; IOS may fall back to version 1.99 (v1 compatibility) depending on release and RSA key state.
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Remediation: Generate an adequate RSA key and pin SSH to version 2.

```
configure terminal
ip domain name <your.domain>
crypto key generate rsa modulus 2048
ip ssh version 2
end
copy running-config startup-config
```

FAIL [HIGH] NIST-SC-12-INTERNAL — Unset default SNMP community names

Framework: NIST SP 800-53 · Control: no_default_snmp_community (internal semantic mapping)

No SNMP community string may use a well-known default value such as 'public' or 'private'.

Why: snmp_communities.name=['public', 'private'] (required: contains_none ['public', 'private'])

Evidence

snmp_communities	L27: snmp-server community public R0
------------------	--------------------------------------

Remediation: Remove default community strings and replace them with unique, non-guessable values restricted by an ACL.

```
configure terminal
no snmp-server community public
no snmp-server community private
access-list 99 permit <nms-subnet> <wildcard>
snmp-server community <unique-string> R0 99
end
copy running-config startup-config
```

FAIL [HIGH] NIST-SC-12-RW-INTERNAL — Do not configure read-write SNMP communities

Framework: NIST SP 800-53 · Control: no_write_snmp_community (internal semantic mapping)

No SNMP v1/v2c community may grant read-write access.

Why: snmp_communities.access=['ro', 'rw'] (required: contains_none ['rw'])

Evidence

snmp_communities	L27: snmp-server community public R0
------------------	--------------------------------------

Remediation: Remove every read-write community; use read-only with an ACL, or SNMPv3 for write access.

```
configure terminal
no snmp-server community <rw-string> RW
snmp-server community <new-ro-string> R0 99
end
copy running-config startup-config
```

FAIL [HIGH] NIST-AC-17 — Restrict VTY transport to secure protocols only

Framework: NIST SP 800-53 · Control: AC-17 (verified reference)

VTY lines must accept SSH only. Telnet and other legacy cleartext transports must be disabled.

Why: telnet_enabled=True (required: is_false); vty_transport_input=['acercon', 'lat', 'pad', 'rlogin', 'ssh', 'telnet', 'udptn'] (required: subset_of ['ssh'])

Evidence

telnet_enabled	L41: transport input telnet
vty_transport_input	L41: transport input telnet

Remediation: Restrict every VTY line to the SSH transport.

```
configure terminal
line vty 0 15
transport input ssh
end
copy running-config startup-config
```

FAIL [HIGH] NIST-AC-3 — Restrict management access by source IP

Framework: NIST SP 800-53 · Control: AC-3 (verified reference)

An ACL/filter must be applied to restrict administrative access to authorized source subnets.

Why: management_acl_applied=False (required: is_true)

Evidence

management_acl_applied	<absent> 2 'line vty' block(s) have no inbound 'access-class', so remote management is reachable from any source address.
------------------------	---

Remediation: Permit only the management network, and apply the list inbound on every VTY line.

```
configure terminal
access-list 99 remark MANAGEMENT-HOSTS
access-list 99 permit 10.20.30.0 0.0.0.255
line vty 0 15
access-class 99 in
end
copy running-config startup-config
```

FAIL [HIGH] NIST-IA-5 — Store privileged credential as hash and encrypt passwords

Framework: NIST SP 800-53 · Control: IA-5 (verified reference)

The privileged credential must be stored as a one-way hash, and stored passwords elsewhere in the configuration must not be left in plaintext.

Why: enable_secret_set=False (required: is_true); password_encryption=False (required: is_true)

Evidence

enable_secret_set	<absent> No 'enable secret' statement present. IOS always writes this command to the running-config when configured, so its absence means no enable secret is set.
password_encryption	<absent> No 'service password-encryption' statement present. The feature is disabled by default and is written to the running-config when enabled, so absence means disabled.

Remediation: Configure a hashed enable secret and turn on password encryption; remove the legacy enable password.

```
configure terminal
enable secret <strong-password>
no enable password
service password-encryption
end
copy running-config startup-config
```

FAIL [MEDIUM] NIST-AC-12 — Set VTY session idle timeout

Framework: NIST SP 800-53 · Control: AC-12 (verified reference)

Idle VTY sessions must be torn down automatically within 10 minutes (600 seconds) or less.

Why: vty_exec_timeout_seconds=0 (required: greater_than 0)

Evidence

vtty_exec_timeout_seconds	L40: exec-timeout 0 0
---------------------------	-----------------------

Remediation: Set an idle timeout of 10 minutes or less on every VTY line.

```
configure terminal
line vty 0 15
exec-timeout 10 0
end
copy running-config startup-config
```

FAIL [MEDIUM] NIST-AC-2 — Enable AAA Authentication

Framework: NIST SP 800-53 · Control: AC-2 (verified reference)

AAA must be enabled so that authentication, authorization and accounting for device access are centrally defined rather than falling back to line-local passwords.

Why: aaa_enabled=False (required: is_true)

Evidence

aaa_enabled	<absent> No 'aaa new-model' statement present. IOS writes this command to the running-config when enabled, so its absence means AAA is not enabled.
-------------	---

Remediation: Enable the AAA subsystem, then define authentication methods before disconnecting.

```
configure terminal
aaa new-model
end
copy running-config startup-config
```

FAIL [MEDIUM] NIST-AU-2 — Configure a logging destination

Framework: NIST SP 800-53 · Control: AU-2 (verified reference)

Logging must be enabled and must have at least one destination: a remote syslog host, a sized local buffer, or both.

Why: logging_enabled=False (required: is_true); logging_hosts=[] (required: is_not_empty); logging_buffered=False (required: is_true)

Evidence

logging_enabled	<absent> No 'logging host' or 'logging buffered' statement present. Both are written to the running-config when configured, so absence means no log destination exists.
logging_hosts	<absent> No 'logging host' statement present; no remote syslog destination is configured.
logging_buffered	<absent> No 'logging buffered' statement present; local log buffering is not configured.

Remediation: Enable logging to a sized local buffer and to a remote syslog collector.

```
configure terminal
logging on
logging buffered 64000 informational
logging host <syslog-server-ip>
logging trap informational
service timestamps log datetime msec show-timezone
end
copy running-config startup-config
```

FAIL [MEDIUM] NIST-AU-8 — Configure at least one NTP server

Framework: NIST SP 800-53 · Control: AU-8 (verified reference)

The device must synchronise its clock with at least one NTP time source.

Why: ntp_servers=[] (required: is_not_empty)

Evidence

ntp_servers	<absent> No 'ntp server' statement present. IOS writes NTP configuration to the running-config, so absence means the clock is not synchronised.
-------------	---

Remediation: Point the device at two or more internal time sources.

```
configure terminal
ntp server 10.20.30.41
ntp server 10.20.30.42
clock timezone UTC 0 0
end
copy running-config startup-config
```

FAIL [MEDIUM] NIST-IA-5-1 — Enforce minimum password length

Framework: NIST SP 800-53 · Control: IA-5(1) (verified reference)

The device must refuse locally configured passwords shorter than eight characters.

Why: password_min_length=0 (required: greater_or_equal 8)

Evidence

password_min_length	<absent> No 'security passwords min-length' statement present. IOS enforces no minimum password length unless this is configured, and writes it when it is.
---------------------	---

Remediation: Enforce a minimum length for every locally configured password.

```
configure terminal
security passwords min-length 8
end
copy running-config startup-config
```

FAIL [MEDIUM] NIST-SC-7 — Disable HTTP management server

Framework: NIST SP 800-53 · Control: SC-7 (verified reference)

The on-box unencrypted HTTP management server must be disabled.

Why: http_server_enabled=True (required: is_false)

Evidence

http_server_enabled	L14: ip http server
---------------------	---------------------

Remediation: Disable the HTTP server; use HTTPS or SSH if a management interface is genuinely required.

```
configure terminal
no ip http server
no ip http secure-server
end
copy running-config startup-config
```

FAIL [LOW] NIST-AC-8 — Configure a login banner

Framework: NIST SP 800-53 · Control: AC-8 (verified reference)

A banner must be displayed before or at login to warn unauthorized users.

Why: login_banner_present=False (required: is_true)

Evidence

login_banner_present	<absent> No 'banner login', 'banner motd' or 'banner exec' statement present. IOS writes banner configuration to the running-config, so absence means no banner is shown.
----------------------	---

Remediation: Set a login banner stating that access is restricted and activity is monitored.

```
configure terminal
banner login ^C
Authorised access only. Activity is logged and monitored.
^C
end
copy running-config startup-config
```

NEEDS REVIEW [HIGH] NIST-SC-13 — Enforce SSH protocol version 2

Framework: NIST SP 800-53 · Control: SC-13 (verified reference)

The device must enforce SSH protocol version 2 for secure administrative management.

Why: No conclusive evidence for ssh_version. No 'ip ssh version' statement found; IOS may fall back to version 1.99 (v1 compatibility) depending on release and RSA key state.

Evidence

ssh_version	<no evidence> No 'ip ssh version' statement found; IOS may fall back to version 1.99 (v1 compatibility) depending on release and RSA key state.
-------------	---

Remediation: Generate an adequate RSA key and pin SSH to version 2.

```
configure terminal
ip domain name <your.domain>
crypto key generate rsa modulus 2048
ip ssh version 2
end
copy running-config startup-config
```

FAIL [HIGH] STIG-MGMT-ACL-INTERNAL — Restrict management access by source IP

Framework: DISA STIG · Control: management_acl (internal semantic mapping)

An ACL/filter must be applied to restrict administrative access to authorized source subnets.

Why: management_acl_applied=False (required: is_true)

Evidence

management_acl_applied	<absent> 2 'line vty' block(s) have no inbound 'access-class', so remote management is reachable from any source address.
------------------------	---

Remediation: Permit only the management network, and apply the list inbound on every VTY line.

```
configure terminal
access-list 99 remark MANAGEMENT-HOSTS
access-list 99 permit 10.20.30.0 0.0.0.255
line vty 0 15
access-class 99 in
end
copy running-config startup-config
```

FAIL [HIGH] STIG-SNMP-DEFAULT-INTERNAL — Unset default SNMP community names

Framework: DISA STIG · Control: no_default_snmp_community (internal semantic mapping)

No SNMP community string may use a well-known default value such as 'public' or 'private'.

Why: snmp_communities.name=['public', 'private'] (required: contains_none ['public', 'private'])

Evidence

snmp_communities	L27: snmp-server community public R0
------------------	--------------------------------------

Remediation: Remove default community strings and replace them with unique, non-guessable values restricted by an ACL.

```
configure terminal
no snmp-server community public
no snmp-server community private
access-list 99 permit <nms-subnet> <wildcard>
snmp-server community <unique-string> R0 99
end
copy running-config startup-config
```

FAIL [HIGH] STIG-SNMP-RW-INTERNAL — Do not configure read-write SNMP communities

Framework: DISA STIG · Control: no_write_snmp_community (internal semantic mapping)

No SNMP v1/v2c community may grant read-write access.

Why: snmp_communities.access=['ro', 'rw'] (required: contains_none ['rw'])

Evidence

snmp_communities	L27: snmp-server community public R0
------------------	--------------------------------------

Remediation: Remove every read-write community; use read-only with an ACL, or SNMPv3 for write access.

```
configure terminal
no snmp-server community <rw-string> RW
snmp-server community <new-ro-string> R0 99
end
copy running-config startup-config
```

FAIL [HIGH] STIG-CCI-000200 — Store privileged credential as hash and encrypt passwords

Framework: DISA STIG · Control: CCI-000200 (verified reference)

The privileged credential must be stored as a one-way hash, and stored passwords elsewhere in the configuration must not be left in plaintext.

Why: enable_secret_set=False (required: is_true); password_encryption=False (required: is_true)

Evidence

enable_secret_set	<absent> No 'enable secret' statement present. IOS always writes this command to the running-config when configured, so its absence means no enable secret is set.
password_encryption	<absent> No 'service password-encryption' statement present. The feature is disabled by default and is written to the running-config when enabled, so absence means disabled.

Remediation: Configure a hashed enable secret and turn on password encryption; remove the legacy enable password.

```
configure terminal
enable secret <strong-password>
no enable password
service password-encryption
end
copy running-config startup-config
```

FAIL [HIGH] STIG-CCI-000366 — Restrict VTY transport to secure protocols only

Framework: DISA STIG · Control: CCI-000366 (verified reference)

VTY lines must accept SSH only. Telnet and other legacy cleartext transports must be disabled.

Why: telnet_enabled=True (required: is_false); vty_transport_input=['acercon', 'lat', 'pad', 'rlogin', 'ssh', 'telnet', 'udptn'] (required: subset_of ['ssh'])

Evidence

telnet_enabled	L41: transport input telnet
vty_transport_input	L41: transport input telnet

Remediation: Restrict every VTY line to the SSH transport.

```
configure terminal
line vty 0 15
transport input ssh
end
copy running-config startup-config
```

FAIL [MEDIUM] STIG-CCI-000015 — Enable AAA Authentication

Framework: DISA STIG · Control: CCI-000015 (verified reference)

AAA must be enabled so that authentication, authorization and accounting for device access are centrally defined rather than falling back to line-local passwords.

Why: aaa_enabled=False (required: is_true)

Evidence

aaa_enabled	<absent> No 'aaa new-model' statement present. IOS writes this command to the running-config when enabled, so its absence means AAA is not enabled.
-------------	---

Remediation: Enable the AAA subsystem, then define authentication methods before disconnecting.

```
configure terminal
aaa new-model
end
copy running-config startup-config
```

FAIL [MEDIUM] STIG-CCI-000057 — Set VTY session idle timeout

Framework: DISA STIG · Control: CCI-000057 (verified reference)

Idle VTY sessions must be torn down automatically within 10 minutes (600 seconds) or less.

Why: vty_exec_timeout_seconds=0 (required: greater_than 0)

Evidence

vtty_exec_timeout_seconds	L40: exec-timeout 0 0
---------------------------	-----------------------

Remediation: Set an idle timeout of 10 minutes or less on every VTY line.

```
configure terminal
line vty 0 15
exec-timeout 10 0
end
copy running-config startup-config
```

FAIL [MEDIUM] STIG-CCI-000130 — Configure a logging destination

Framework: DISA STIG · Control: CCI-000130 (verified reference)

Logging must be enabled and must have at least one destination: a remote syslog host, a sized local buffer, or both.

Why: logging_enabled=False (required: is_true); logging_hosts=[] (required: is_not_empty); logging_buffered=False (required: is_true)

Evidence

logging_enabled	<absent> No 'logging host' or 'logging buffered' statement present. Both are written to the running-config when configured, so absence means no log destination exists.
logging_hosts	<absent> No 'logging host' statement present; no remote syslog destination is configured.
logging_buffered	<absent> No 'logging buffered' statement present; local log buffering is not configured.

Remediation: Enable logging to a sized local buffer and to a remote syslog collector.

```
configure terminal
logging on
logging buffered 64000 informational
logging host <syslog-server-ip>
logging trap informational
service timestamps log datetime msec show-timezone
end
copy running-config startup-config
```


FAIL [MEDIUM] STIG-CCI-000159 — Configure at least one NTP server

Framework: DISA STIG · Control: CCI-000159 (verified reference)

The device must synchronise its clock with at least one NTP time source.

Why: ntp_servers=[] (required: is_not_empty)

Evidence

ntp_servers	<absent> No 'ntp server' statement present. IOS writes NTP configuration to the running-config, so absence means the clock is not synchronised.
-------------	---

Remediation: Point the device at two or more internal time sources.

```
configure terminal
ntp server 10.20.30.41
ntp server 10.20.30.42
clock timezone UTC 0 0
end
copy running-config startup-config
```

FAIL [MEDIUM] STIG-CCI-000200-LEN — Enforce minimum password length

Framework: DISA STIG · Control: CCI-000200 (verified reference)

The device must refuse locally configured passwords shorter than eight characters.

Why: password_min_length=0 (required: greater_or_equal 8)

Evidence

password_min_length	<absent> No 'security passwords min-length' statement present. IOS enforces no minimum password length unless this is configured, and writes it when it is.
---------------------	---

Remediation: Enforce a minimum length for every locally configured password.

```
configure terminal
security passwords min-length 8
end
copy running-config startup-config
```

FAIL [MEDIUM] STIG-CCI-000381 — Disable HTTP management server

Framework: DISA STIG · Control: CCI-000381 (verified reference)

The on-box unencrypted HTTP management server must be disabled.

Why: http_server_enabled=True (required: is_false)

Evidence

http_server_enabled	L14: ip http server
---------------------	---------------------

Remediation: Disable the HTTP server; use HTTPS or SSH if a management interface is genuinely required.

```
configure terminal
no ip http server
no ip http secure-server
end
copy running-config startup-config
```

FAIL [LOW] STIG-CCI-000048 — Configure a login banner

Framework: DISA STIG · Control: CCI-000048 (verified reference)

A banner must be displayed before or at login to warn unauthorized users.

Why: login_banner_present=False (required: is_true)

Evidence

login_banner_present	<absent> No 'banner login', 'banner motd' or 'banner exec' statement present. IOS writes banner configuration to the running-config, so absence means no banner is shown.
----------------------	---

Remediation: Set a login banner stating that access is restricted and activity is monitored.

```
configure terminal
banner login ^C
Authorised access only. Activity is logged and monitored.
^C
end
copy running-config startup-config
```

NEEDS REVIEW [HIGH] STIG-CCI-000068 — Enforce SSH protocol version 2

Framework: DISA STIG · Control: CCI-000068 (verified reference)

The device must enforce SSH protocol version 2 for secure administrative management.

Why: No conclusive evidence for ssh_version. No 'ip ssh version' statement found; IOS may fall back to version 1.99 (v1 compatibility) depending on release and RSA key state.

Evidence

ssh_version	<no evidence> No 'ip ssh version' statement found; IOS may fall back to version 1.99 (v1 compatibility) depending on release and RSA key state.
-------------	---

Remediation: Generate an adequate RSA key and pin SSH to version 2.

```
configure terminal
ip domain name <your.domain>
crypto key generate rsa modulus 2048
ip ssh version 2
end
copy running-config startup-config
```

FAIL [HIGH] ISO-A.5.17-ENC — Store privileged credential as hash and encrypt passwords

Framework: ISO/IEC 27001 · Control: enable_secret_encrypted (internal semantic mapping)

The privileged credential must be stored as a one-way hash, and stored passwords elsewhere in the configuration must not be left in plaintext.

Why: enable_secret_set=False (required: is_true); password_encryption=False (required: is_true)

Evidence

enable_secret_set	<absent> No 'enable secret' statement present. IOS always writes this command to the running-config when configured, so its absence means no enable secret is set.
password_encryption	<absent> No 'service password-encryption' statement present. The feature is disabled by default and is written to the running-config when enabled, so absence means disabled.

Remediation: Configure a hashed enable secret and turn on password encryption; remove the legacy enable password.

```
configure terminal
enable secret <strong-password>
no enable password
service password-encryption
end
copy running-config startup-config
```

FAIL [HIGH] ISO-A.8.22-ACL — Restrict management access by source IP

Framework: ISO/IEC 27001 · Control: management_acl (internal semantic mapping)

An ACL/filter must be applied to restrict administrative access to authorized source subnets.

Why: management_acl_applied=False (required: is_true)

Evidence

management_acl_applied	<absent> 2 'line vty' block(s) have no inbound 'access-class', so remote management is reachable from any source address.
------------------------	---

Remediation: Permit only the management network, and apply the list inbound on every VTY line.

```
configure terminal
access-list 99 remark MANAGEMENT-HOSTS
access-list 99 permit 10.20.30.0 0.0.0.255
line vty 0 15
access-class 99 in
end
copy running-config startup-config
```

FAIL [HIGH] ISO-A.8.24-DEFAULT — Unset default SNMP community names

Framework: ISO/IEC 27001 · Control: no_default_snmp_community (internal semantic mapping)

No SNMP community string may use a well-known default value such as 'public' or 'private'.

Why: snmp_communities.name=['public', 'private'] (required: contains_none ['public', 'private'])

Evidence

snmp_communities	L27: snmp-server community public R0
------------------	--------------------------------------

Remediation: Remove default community strings and replace them with unique, non-guessable values restricted by an ACL.

```
configure terminal
no snmp-server community public
no snmp-server community private
access-list 99 permit <nms-subnet> <wildcard>
snmp-server community <unique-string> R0 99
end
copy running-config startup-config
```

FAIL [HIGH] ISO-A.8.24-RW — Do not configure read-write SNMP communities

Framework: ISO/IEC 27001 · Control: no_write_snmp_community (internal semantic mapping)

No SNMP v1/v2c community may grant read-write access.

Why: snmp_communities.access=['ro', 'rw'] (required: contains_none ['rw'])

Evidence

snmp_communities	L27: snmp-server community public R0
------------------	--------------------------------------

Remediation: Remove every read-write community; use read-only with an ACL, or SNMPv3 for write access.

```
configure terminal
no snmp-server community <rw-string> RW
snmp-server community <new-ro-string> R0 99
end
copy running-config startup-config
```

FAIL [HIGH] ISO-A.8.20 — Restrict VTY transport to secure protocols only

Framework: ISO/IEC 27001 · Control: A.8.20 (verified reference)

VTY lines must accept SSH only. Telnet and other legacy cleartext transports must be disabled.

Why: telnet_enabled=True (required: is_false); vty_transport_input=['acercon', 'lat', 'pad', 'rlogin', 'ssh', 'telnet', 'udptn'] (required: subset_of ['ssh'])

Evidence

telnet_enabled	L41: transport input telnet
vty_transport_input	L41: transport input telnet

Remediation: Restrict every VTY line to the SSH transport.

```
configure terminal
line vty 0 15
transport input ssh
end
copy running-config startup-config
```

FAIL [MEDIUM] ISO-A.8.10-NTP — Configure at least one NTP server

Framework: ISO/IEC 27001 · Control: ntp_configured (internal semantic mapping)

The device must synchronise its clock with at least one NTP time source.

Why: ntp_servers=[] (required: is_not_empty)

Evidence

ntp_servers	<absent> No 'ntp server' statement present. IOS writes NTP configuration to the running-config, so absence means the clock is not synchronised.
-------------	---

Remediation: Point the device at two or more internal time sources.

```
configure terminal
ntp server 10.20.30.41
ntp server 10.20.30.42
clock timezone UTC 0 0
end
copy running-config startup-config
```

FAIL [MEDIUM] ISO-A.8.22-HTTP — Disable HTTP management server

Framework: ISO/IEC 27001 · Control: http_server_disabled (internal semantic mapping)

The on-box unencrypted HTTP management server must be disabled.

Why: http_server_enabled=True (required: is_false)

Evidence

http_server_enabled	L14: ip http server
---------------------	---------------------

Remediation: Disable the HTTP server; use HTTPS or SSH if a management interface is genuinely required.

```
configure terminal
no ip http server
no ip http secure-server
end
copy running-config startup-config
```

FAIL [MEDIUM] ISO-A.5.17 — Enforce minimum password length

Framework: ISO/IEC 27001 · Control: A.5.17 (verified reference)

The device must refuse locally configured passwords shorter than eight characters.

Why: password_min_length=0 (required: greater_or_equal 8)

Evidence

password_min_length	<absent> No 'security passwords min-length' statement present. IOS enforces no minimum password length unless this is configured, and writes it when it is.
---------------------	---

Remediation: Enforce a minimum length for every locally configured password.

```
configure terminal
security passwords min-length 8
end
copy running-config startup-config
```

FAIL [MEDIUM] ISO-A.8.10 — Configure a logging destination

Framework: ISO/IEC 27001 · Control: A.8.10 (verified reference)

Logging must be enabled and must have at least one destination: a remote syslog host, a sized local buffer, or both.

Why: logging_enabled=False (required: is_true); logging_hosts=[] (required: is_not_empty); logging_buffered=False (required: is_true)

Evidence

logging_enabled	<absent> No 'logging host' or 'logging buffered' statement present. Both are written to the running-config when configured, so absence means no log destination exists.
logging_hosts	<absent> No 'logging host' statement present; no remote syslog destination is configured.
logging_buffered	<absent> No 'logging buffered' statement present; local log buffering is not configured.

Remediation: Enable logging to a sized local buffer and to a remote syslog collector.

```
configure terminal
logging on
logging buffered 64000 informational
logging host <syslog-server-ip>
logging trap informational
service timestamps log datetime msec show-timezone
end
copy running-config startup-config
```

FAIL [MEDIUM] ISO-A.8.19 — Set VTY session idle timeout

Framework: ISO/IEC 27001 · Control: A.8.19 (verified reference)

Idle VTY sessions must be torn down automatically within 10 minutes (600 seconds) or less.

Why: vty_exec_timeout_seconds=0 (required: greater_than 0)

Evidence

vty_exec_timeout_seconds	L40: exec-timeout 0 0
--------------------------	-----------------------

Remediation: Set an idle timeout of 10 minutes or less on every VTY line.

```
configure terminal
line vty 0 15
exec-timeout 10 0
end
copy running-config startup-config
```

FAIL [MEDIUM] ISO-A.8.2 — Enable AAA Authentication

Framework: ISO/IEC 27001 · Control: A.8.2 (verified reference)

AAA must be enabled so that authentication, authorization and accounting for device access are centrally defined rather than falling back to line-local passwords.

Why: aaa_enabled=False (required: is_true)

Evidence

aaa_enabled	<absent> No 'aaa new-model' statement present. IOS writes this command to the running-config when enabled, so its absence means AAA is not enabled.
-------------	---

Remediation: Enable the AAA subsystem, then define authentication methods before disconnecting.

```
configure terminal
aaa new-model
end
copy running-config startup-config
```

FAIL [LOW] ISO-A.8.20-BANNER — Configure a login banner

Framework: ISO/IEC 27001 · Control: login_banner (internal semantic mapping)

A banner must be displayed before or at login to warn unauthorized users.

Why: login_banner_present=False (required: is_true)

Evidence

login_banner_present	<absent> No 'banner login', 'banner motd' or 'banner exec' statement present. IOS writes banner configuration to the running-config, so absence means no banner is shown.
----------------------	---

Remediation: Set a login banner stating that access is restricted and activity is monitored.

```
configure terminal
banner login ^C
Authorised access only. Activity is logged and monitored.
^C
end
copy running-config startup-config
```

NEEDS_REVIEW [HIGH] ISO-A.8.20-SSH — Enforce SSH protocol version 2

Framework: ISO/IEC 27001 · Control: ssh_version_2 (internal semantic mapping)

The device must enforce SSH protocol version 2 for secure administrative management.

Why: No conclusive evidence for ssh_version. No 'ip ssh version' statement found; IOS may fall back to version 1.99 (v1 compatibility) depending on release and RSA key state.

Evidence

ssh_version	<no evidence> No 'ip ssh version' statement found; IOS may fall back to version 1.99 (v1 compatibility) depending on release and RSA key state.
-------------	---

Remediation: Generate an adequate RSA key and pin SSH to version 2.

```
configure terminal
ip domain name <your.domain>
crypto key generate rsa modulus 2048
ip ssh version 2
end
copy running-config startup-config
```

Notes

- NEEDS_REVIEW means the configuration carried no conclusive evidence either way. It is not a pass, and it is not a failure: it is an escalation to a human.
- Every line number cited above refers to the ingested configuration file identified under Source and Provenance.
- 4 control result(s) on this device could not be decided from the configuration and require review.
- No hardware serial number was available for this device. Supply the output of the vendor's show-version command alongside the configuration to record one.